

KAMTA NLONFO JORDAN

Physicist specialised in Environmental Physics

Climate modelling | Climate variability | Data analysis | Teaching

PERSONAL INFORMATION

Phone: +237 652 41 52 27 / +39 351 469 7270
E-mail: Jordankamta35@gmail.com / Jkamta_n@ictp.it
Address: Via dei Bonomo 2, 34126 Trieste, Italy
Languages: French (native), English (B2)

PROFESSIONAL PROFILE

Environmental physicist with a strong interest in the study of complex systems, ocean-atmosphere interactions and climate variability. My expertise is built on the command of advanced numerical tools applied to the study of the climate system and its physical mechanisms. My academic path has been enriched by several international schools and high-level training programmes, which have consolidated my methodological foundations in environmental physics and climate modelling, while giving me a rare competence in the coupling of oceanographic and atmospheric analyses, as well as in artificial intelligence applied to Earth sciences. Determined to contribute to high-level research and to climate adaptation strategies, I intend to pursue a PhD and to place my expertise at the service of decision-support tools for the resilience of populations.

TECHNICAL SKILLS

- **Modelling and analysis of environmental systems**
Expertise in the use of the regional climate model REGCM5 and the ocean model CROCO. Numerical simulation of regional atmospheric systems and study of atmospheric stability. Study of extreme events and their impacts.
- **Data management and processing**
Retrieval, extraction and processing of climate, oceanographic and satellite data. Analysis and visualisation of multi-format data: NetCDF, HDF5, GRIB, CSV, GeoTIFF, as well as reanalysis and satellite observation databases. Application of advanced statistical methods: EOF/PCA, trend detection, anomaly computation, multiple correlations and regressions, spectral analysis (Morlet wavelets, FFT), model cross-validation.
- **Scientific computing, AI and GIS**
Advanced command of PYTHON (xarray, NumPy, Pandas, Matplotlib, Cartopy), FORTRAN, NCL, CDO. Foundations in machine learning applied to Earth system sciences (CNN/GNN/Transformer architectures, ANEMOI framework). Experience with geographic information systems: ARCGIS, QGIS. Additional skills in web development (HTML, CSS, JavaScript).

PROFESSIONAL EXPERIENCE

- **Physics teacher** – HEI-ISTIC (Higher School of Engineering – Higher Institute of Industrial and Commercial Techniques), Bangangté 2024 – 2026
Courses taught: particle mechanics, fluid mechanics, thermodynamics, statics.
- **Teacher** – SONNANG Institute, Dschang 2024 – 2026
Courses taught: thermodynamics, rigid-body and particle mechanics.
- **Tutor and scientific mentor** – Intellect Group & private tuition 2021 – 2026
Examination preparation (Science Baccalaureate, competitive entrance examinations).

EDUCATION

- **ICTP Postgraduate Diploma in Earth System Physics**, Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy Sept. 2026 – Aug. 2027
Ongoing pre-PhD postgraduate programme in Earth System Physics. Core training covers wave physics, physics of the Earth for geohazards, atmospheric thermodynamics, climate physics, geophysical fluid dynamics, numerical methods, Linux and scientific Python, followed by advanced coursework and a research dissertation.
- **Master's Degree in Environmental Physics**, University of Dschang 2024 – 2025
Grade: Highest Honours.
Master's thesis: "Study of Atmospheric Static Stability over Central Africa Using the RegCM5 Model"
- **Bachelor's Degree in Physics** – Specialisation in Environmental Physics, University of Dschang 2022 – 2023
Grade: Honours .
- **Science Baccalaureate** – Mathematics/Physics stream, Rural Bilingual High School of Mbouda 2019 – 2020
Grade: Honours.

ADDITIONAL TRAINING AND CONFERENCES

- **Certificate of Achievement – Fundamentals of Ocean Prediction** 27 July – 14 September 2026
ICMPA-UNESCO Chair, Gulf of Guinea Ocean Sciences Summer School (GGOSSS), University of Abomey-Calavi & Ocean Prediction DCC – FutureLearn

OPERA MOOC (Set 1, 7 weeks, 4 h/week) completed with an overall score of **96%**. Training covering end-to-end ocean forecasting systems: ocean dynamics, numerical models, observations and data assimilation, upstream data sources, forecast interpretation and downstream applications for maritime safety, fisheries and coastal risk management.

- **9th Biennial African School of Fundamental Physics and Applications (ASP2026)** 5 – 19 July 2026
ICTP (Abdus Salam International Centre for Theoretical Physics) – Nairobi, Kenya
Intensive two-week training in fundamental and applied physics: high-energy physics, cosmology, condensed matter physics, astroparticle physics and societal applications. Organised in partnership with Brookhaven National Laboratory and the University of Nairobi.
- **Geodesy for a Resilient Africa Summer School** 29 June – 3 July 2026
AIMS-Cameroon, Limbé, Cameroon – International Association of Geodesy (IAG)
Intensive training: satellite data processing and interpretation, geodetic reference frames and vertical datums, sea-level monitoring and coastal dynamics, data-driven climate risk assessment.
- **Machine Learning for Earth Systems Modelling – Course 1: Foundations and New Frontiers** 2026
ECMWF, Destination Earth (DestinE) Initiative, European Commission
Fundamentals of machine learning applied to Earth system modelling: digital twins, AI ethics and regulation, future perspectives.
- **Machine Learning for Earth Systems Modelling – Course 2: Architectures, Data and Prediction** 2026
ECMWF, Destination Earth (DestinE) Initiative, European Commission
Neural architectures for atmospheric dynamics (CNN, GNN, Transformers), data management and computing infrastructure, AI-based forecasting systems (AIFS) and the European ANEMOI framework.
- **First International Edition on Modeling and Control of Complex Systems in Life Sciences and Physical Sciences** 24 – 28 November 2025
International conference on advanced modelling techniques and control methods for complex systems.
- **Gulf of Guinea Summer School in Ocean Sciences (GGOSSS)** 25 – 29 August 2025
Summer school on “Oceanographic Expertise and Collaboration for Sustainable Development in the Gulf of Guinea”. Hands-on training: collection and processing of in-situ data, satellite observations, use of regional ocean models and analysis of their outputs.
- **International Scientific School: Simulation Methods in Materials Physics** 21 – 24 May 2025
University of Dschang & Le Mans University
International training on numerical and simulation methods applied to complex physical systems and materials.
- **PWF School on Climate Modelling** 2023
ICTP in collaboration with the University of Dschang
Introduction to the REGCM models; data processing with PYTHON; bias correction, installation and handling of simulation chains.
- **Training on Environmental Professions** 2024
IFDD – International Organisation of La Francophonie
Modules: water access, management and treatment; sustainable energy; sustainable urbanisation and mobility; biotechnology and environmental innovation.
- **Green Soldiers – Climate Life Movement** May 2024
Yaoundé, Cameroon
Climate awareness and action focused on youth engagement and digital tools for environmental activism; advocacy and science communication.
- **Environmental Modelling Seminar** 2024
Polytechnique Montréal & University of Dschang
3D gravimetric interpretation, applied GIS (ARCGIS, QGIS, OASIS MONTAJ), applications in geothermal energy and the environment.

PUBLICATIONS

- “Investigating the Atmospheric Stability over Central Africa Using a Regional Climate Model” (in preparation)

INTERESTS

- Football and endurance running

REFERENCES

Available upon request.